

A

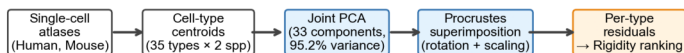
1. Data

2. Aggregate

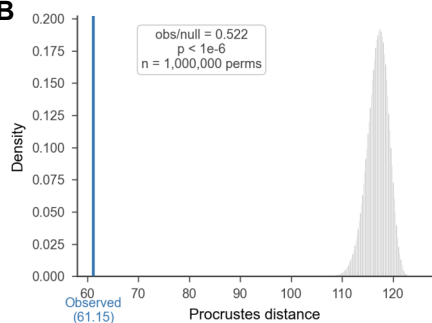
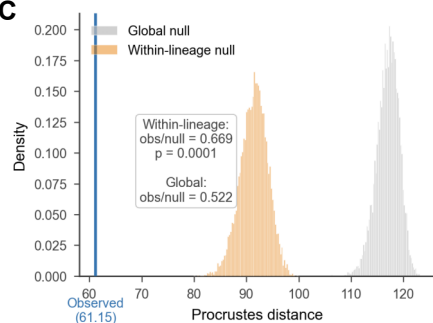
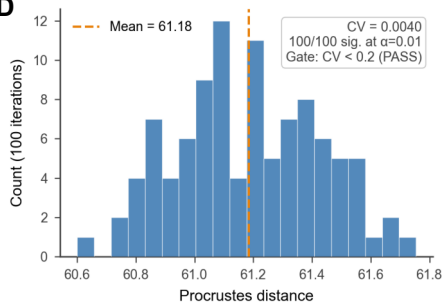
3. Embed

4. Align

5. Quantify



Permutation test: shuffle cell type labels → null distribution → $p < 1e-6$ (1M permutations)

B**C****D****E**